Progress Report: U-Net Extensions for ECG Time-Series Classification

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December 16, 2025

1 Scope of This Report

The report contains progress regarding the following aspects of the project:

- extending the evaluation pipeline,
- adding a ResNet-based baseline,
- improving metric consistency and reproducibility,
- and analyzing performance using capacity-aware comparisons.

2 Experimental Pipeline Improvements

2.1 Reproducibility

To ensure reproducibility and to guarantee the same results across multiple runs, a global random seed was introduced and consistently applied across:

- Python random number generation,
- NumPy,
- PyTorch (CPU and CUDA backends).

2.2 Validation Split and Model Selection

The original training set is now split into a **Train set** and a **Test set** via a standard 80-20 split. During training, the best model checkpoint is selected based on **validation accuracy**. All reported test metrics are computed using this best-validation checkpoint, preventing information leakage from the test set.

2.3 Unified Evaluation Metrics

All trained models are evaluated on the test set using the same metrics:

- overall accuracy,
- balanced accuracy,
- **macro-F1 score**.

Due to strong class imbalance in ECG5000, macro-F1 is used as the **primary ranking metric**, as it weights all classes equally.

3 Addition of a ResNet1D Baseline

To contextualize the U-Net-based results, a 1D ResNet architecture was added as a baseline. This model:

- operates directly on raw ECG time series,
- uses residual connections common in time-series and signal processing literature,
- is trained using the same data splits, optimizer settings, and training protocol as the U-Net variants.

To ensure fair comparison, results are also analyzed under **parameter-matched conditions**, allowing comparisons between models with similar capacity (within ±10% parameter count).

4 Updated Quantitative Results

4.1 Parameter Count vs. Macro-F1

Figure 1 shows macro-F1 scores plotted against the number of trainable parameters for all runs.

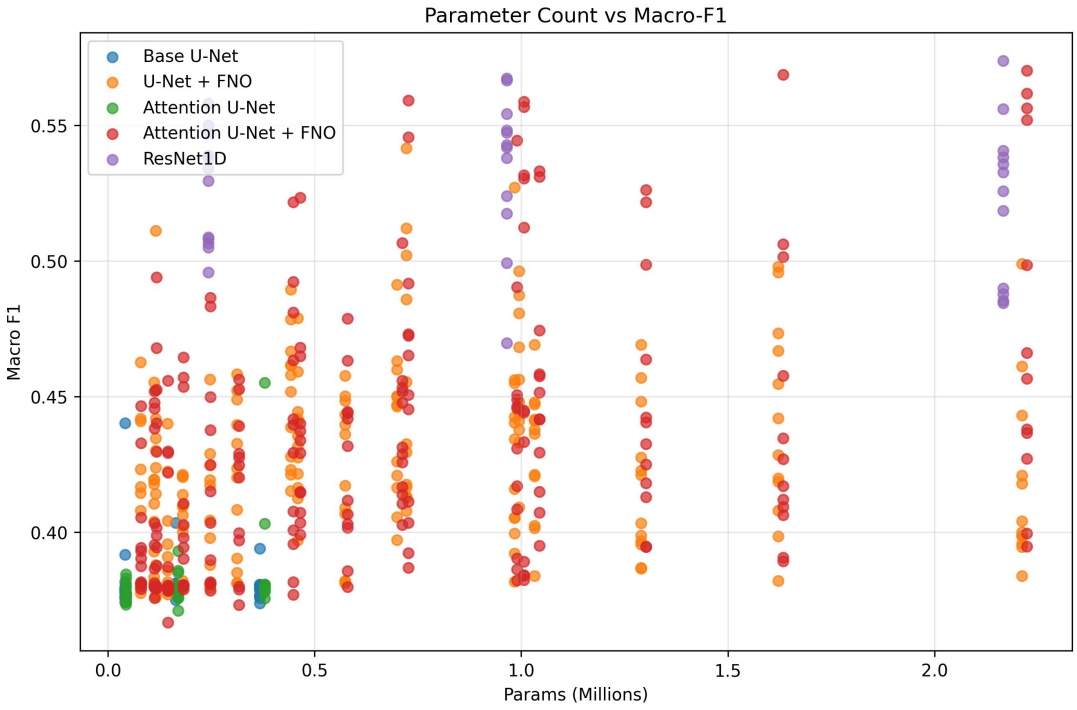


Figure 1: Parameter count vs. macro-F1 across all runs (updated).

This visualization highlights that:

- higher capacity does not automatically imply better macro-F1,
- U-Net + FNO and Attention U-Net + FNO models often achieve competitive macro-F1 at comparable parameter budgets,
- the ResNet1D baseline achieves similar macro-F1 to the strongest U-Net variants at higher parameter counts.

4.2 Top-5 Performance per Model Family

Figure 2 summarizes the distribution of the top-5 macro-F1 scores for each model family.

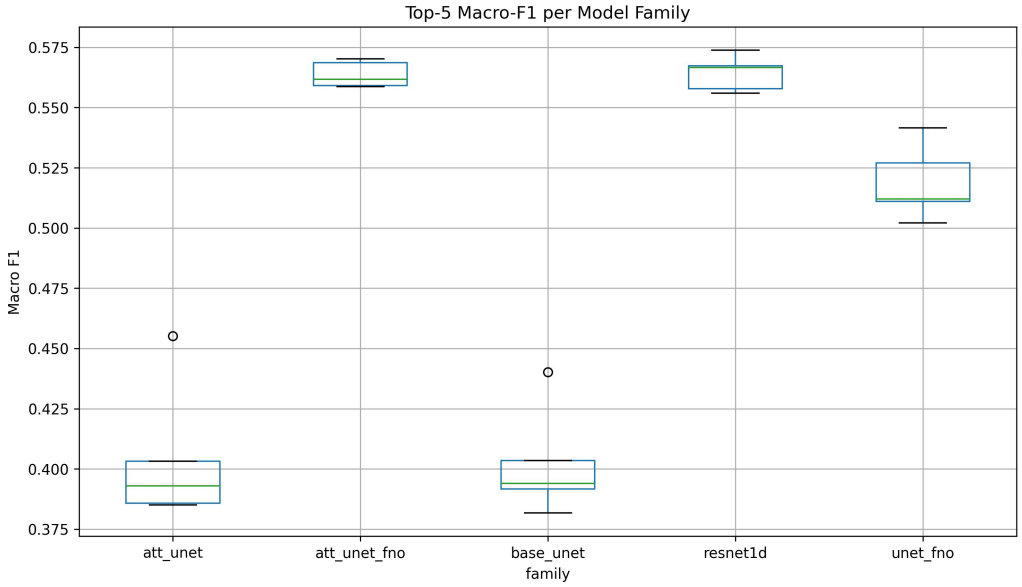


Figure 2: Top-5 macro-F1 per model family (updated).

The results indicate:

- significant variance within each family,
- consistently higher best-case macro-F1 for FNO-augmented models,
- ResNet1D showing strong median performance. It is consistently on par with Attention U-net + FNO models with similar parameter counts.

5 Confusion Matrix Analysis

Figure 3 presents confusion matrices for the best-performing model of each family, selected by macro-F1.

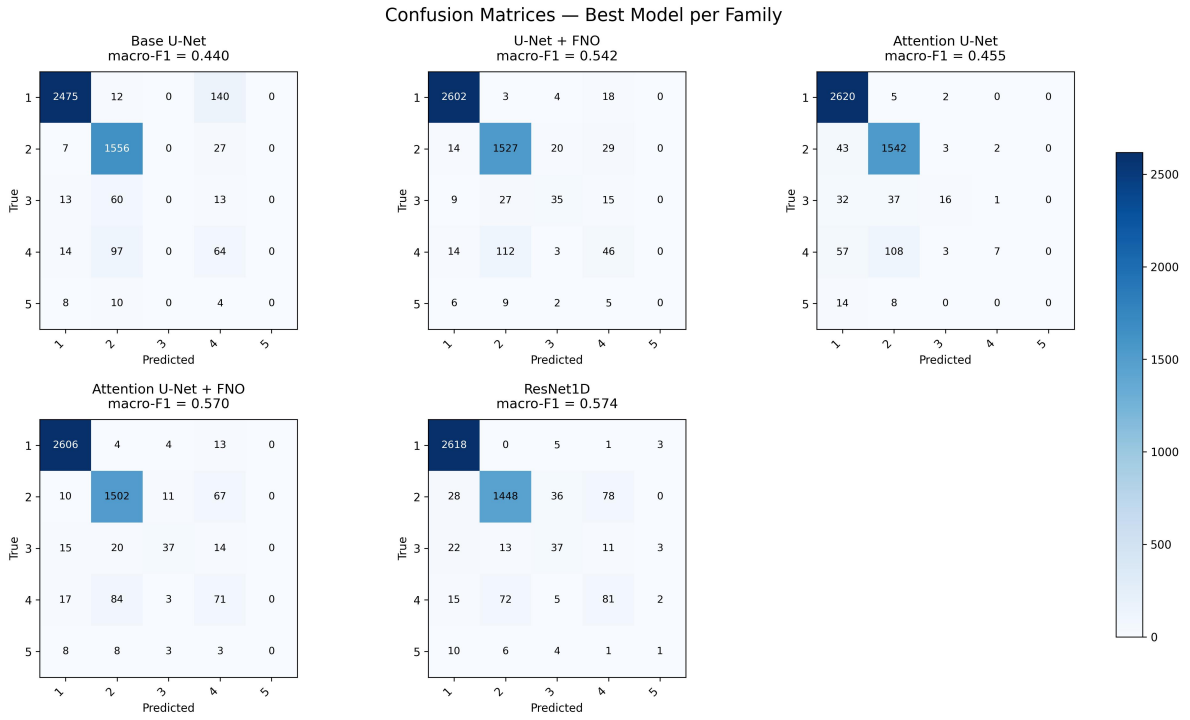


Figure 3: Confusion matrices (counts) for the best model of each family, including the ResNet1D baseline.

Across all architectures, the following trends are observed:

- near-perfect classification for majority classes,
- substantial confusion among minority classes,
- modest but consistent improvements for minority classes when FNO blocks are included.

6 Per-Class F1 Comparison

Figure 4 shows per-class F1 scores for the best model of each family.

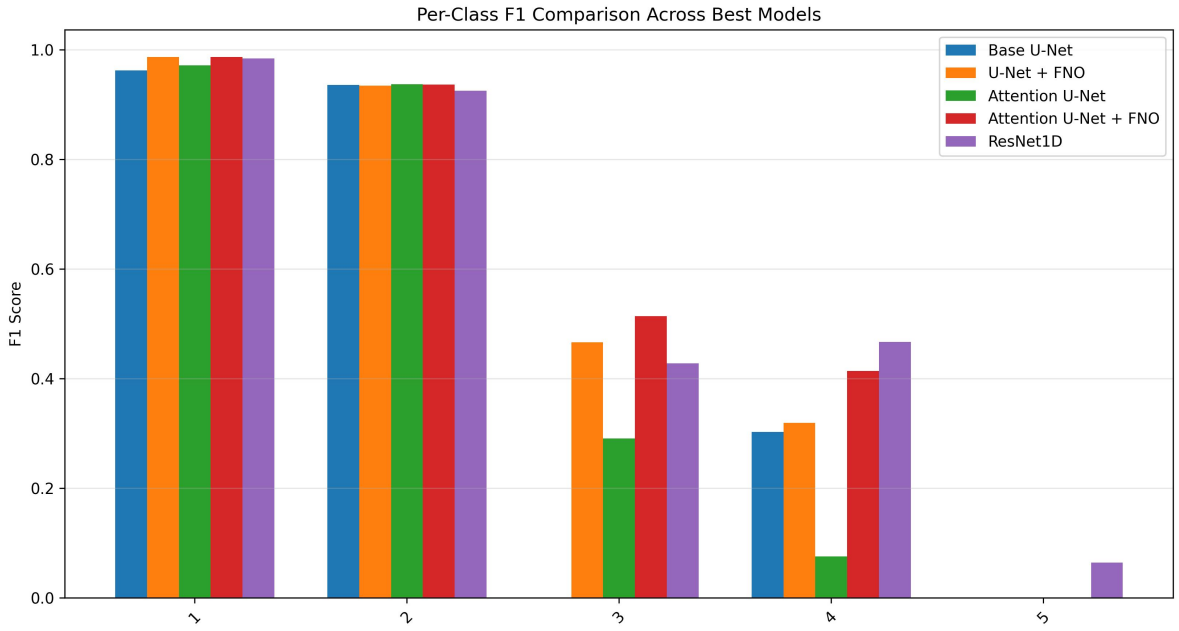


Figure 4: Per-class F1 comparison for best models across all families.

This analysis confirms that:

- macro-F1 improvements stem primarily from better handling of minority classes,
- FNO-augmented models consistently outperform their non-FNO counterparts on under-represented classes,
- ResNet1D achieves competitive performance but does not systematically outperform U-Net variants.

7 Summary of Progress

Since the previous report, the following key advancements were achieved:

- a fully reproducible training and evaluation pipeline,
- consistent validation-based model selection,
- integration of a ResNet1D baseline,
- capacity-aware performance analysis,
- comprehensive qualitative and quantitative evaluation.

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